Zidovex LN
(LAMIVUDINE 150MG, ZIDOVUDINE 300MG, NEVIRAPINE 200MG TABLETS)

Product description

White, capsule shaped, biconvex film coated tablets debossed with “C” on one side and "72"on other side.
Each tablet contains lamivudine 150mg, zidovudine 300mg, and nevirapine 200mg.

Pharmacodynamics

Lamivudine

Lamivudine, a synthetic antiretroviral agent, is a nucleoside reverse transcriptase inhibitor. The drug is a dideoxynucleoside reverse transcriptase inhibitor. Lamivudine is the negative enantiomer of a dideoxy analog of cytidine and is structurally similar to zalcitabine. Lamivudine differs structurally from zalcitabine in that the 3’-carbon of the ribose ring is replaced with sulfur, forming an oxathiolane ring. The absence of a free 3’-hydroxyl group on the oxathiolane ring results in the inability of lamivudine to form phosphodiester linkages at this position.

Both the positive and negative enantiomers of 2’3’-dideoxy, 3’-thiacytidine exhibit antiviral activity in vitro, but lamivudine appears to exhibit greater antiviral activity and to be considerably less cytotoxic than the positive enantiomer.

Following conversion to a pharmacologically active metabolite, lamivudine apparently inhibits replication of human retroviruses by interfering with viral RNA-directed DNA polymerase (reverse transcriptase). Lamivudine therefore exerts a virustatic effect against retroviruses by acting as a reverse transcriptase inhibitor.

The antiviral activity of lamivudine appears to depend on intracellular conversion of the drug to a 5’-triphosphate metabolite; thus, 2’, 3’-dideoxy, 3’-thiacytidine-5’-triphosphate (3TC-TP) and not unchanged lamivudine appears to be the pharmacologically active form of the drug. 3TC-TP is a structural analog of deoxycytidine triphosphate (dC-TP), the natural substrate for reverse transcriptase (viral RNA-directed DNA polymerase). Although other mechanisms may be involved in the antiretroviral activity of the drug, 3TC-TP appears to compete with naturally occurring dC-TP for incorporation into viral DNA by reverse transcriptase. Following incorporation of 3TC-TP into the viral DNA chain instead of dC-TP, viral DNA synthesis is terminated prematurely because the absence of a 3’-hydroxy group on the oxathiolane ring prevents further 5’ to 3’ phosphodiester linkages.

Zidovudine

Zidovudine is an antiviral agent which is highly active *in vitro* against retroviruses including HIV.

Zidovudine is phosphorylated in both infected and uninfected cells to the monophosphate (MP) derivative by cellular thymidine kinase. Subsequent phosphorylation of zidovudine -MP to the diphosphate (DP), and then the triphosphate (TP) derivative is catalysed by cellular thymidylate kinase and non-specific kinases respectively. Zidovudine -TP acts as an inhibitor of and substrate for the viral reverse transcriptase. The formation of further proviral DNA is blocked by incorporation of zidovudine -MP into the chain and subsequent chain termination. Competition by zidovudine -TP for HIV reverse transcriptase is approximately 100-fold greater than for cellular DNA polymerase alpha.

Nevirapine

At the cellular level, nevirapine is selective for HIV-1 reverse transcriptase over other cellular enzymes including human DNA polymerases, calf thymus DNA polymerases and human plasma rennin. In additional studies, nevirapine did not affect human plasma renin activity over a wide range of concentrations studied (5.1 pM to 50 pM). Nevirapine has had no observable cytotoxic effects on cells in general and on cells involved in the immune system in particular.

Nevirapine acts by reversibly inhibiting the activity of HIV-1 reverse transcriptase, an enzyme which directs the polymerization of DNA from viral RNA, a necessary component for HIV-1 replication. For HIV-1 as well as other retroviruses, the viral polymerase has a binding site for nucleoside triphosphates, a site for binding the RNA template primer, and catalytic sites for both the polymerase and ribonuclease H reactions.

The inhibition of the reverse transcriptase directed polymerization of DNA from viral RNA has been an important therapeutic target for the treatment of HIV-1 infection, which was initially reported with the nucleoside analogue zidovudine. Unlike the nucleoside analogues, nevirapine binds directly to the reverse transcriptase at amino acid residues 181 and 188. This site is close to, but not directly at, the polymerase catalytic site on the large subunit of the heterodimeric reverse transcriptase. The binding of nevirapine to reverse transcriptase occurs primarily through hydrophobic interactions to a pocket formed by seven strands. Nevirapine appears to block reverse transcriptase polymerase by altering the position of critical amino acids within the enzymes catalytic site. As a result, the rate of the chemical reaction catalyzed by the reverse transcriptase is significantly slowed. Analysis of kinetic data yielded a dissociation constant (ki) of 217±24 nM for the interaction of Nevirapine with reverse transcriptase. Inhibition is noncompetitive since nevirapine has no effect on the binding of substrate to reverse transcriptase. Nevirapine does not exhibit activity against other viral polymerases, including HIV-2 and simian immunodeficiency virus reverse transcriptase.

Pharmacokinetics

Lamivudine

The pharmacokinetic properties of lamivudine have been studied in asymptomatic, HIV-infected adult patients after administration of single intravenous (IV) doses ranging from 0.25 to 8 mg/kg, as well as single and multiple (b.i.d. regimen) oral doses ranging from 0.25 to 10 mg/kg.

Absorption and Bioavailability

Lamivudine was rapidly absorbed after oral administration in HIV-infected patients. Absolute bioavailability in 12 adult patients was 86% ± 16% (mean ± SD) for the tablet and 87% ± 13% for the oral solution. After oral administration of 2 mg/kg twice a day to 9 adults with HIV, the peak serum lamivudine concentration (C max) was 1.5 ± 0.5 µg/mL (mean ± SD). The area under the plasma concentration versus time curve (AUC) and C max increased in proportion to oral dose over the range from 0.25 to 10 mg/kg. An investigational 25-mg dosage form of lamivudine was administered orally to 12 asymptomatic, HIV-infected patients on 2 occasions, once in the fasted state and once with food (1099 kcal; 75 grams fat, 34 grams protein, 72 grams carbohydrate). Absorption of lamivudine was slower in the fed state (T max: 3.2 ± 1.3 hours) compared with the fasted state (T max : 0.9 ± 0.3 hours); C max in the fed state was 40% ± 23% (mean ± SD) lower than in the fasted state. There was no significant difference in systemic exposure (AUCα) the fed and fasted states; therefore, lamivudine may be administered with or without food. The accumulation ratio of lamivudine in HIV-positive asymptomatic adults with normal renal function was 1.50 following 15 days of oral administration of 2 mg / kg b.i.d.

Distribution

The apparent volume of distribution after IV administration of lamivudine to 20 patients was 1.3 ± 0.4 L/kg, suggesting that lamivudine distributes in extra vascular spaces. Volume of distribution was independent of dose and did not correlate with body weight. Binding of lamivudine to human plasma proteins is low (<36%). In vitro studies showed that, over the concentration range of 0.1 to 100 µg/mL, the amount of lamivudine associated with erythrocytes ranged from 53% to 57% and was independent of concentration.

Metabolism

Metabolism of lamivudine is a minor route of elimination. In man, the only known metabolite of lamivudine is the trans-sulfoxide metabolite. Within 12 hours after a single oral dose of lamivudine in 6 HIV-infected adults, 5.2% ± 1.4% (mean ± SD) of the dose was excreted as the trans-sulfoxide metabolite in the urine. Serum concentrations of this metabolite have not been determined.

Elimination

The majority of lamivudine is eliminated unchanged in urine. In 20 patients given a single IV dose, renal clearance was 0.22 ± 0.06 L/hr*kg (mean ± SD), representing 71% ± 16% (mean ± SD) of total clearance of lamivudine.

In most single-dose studies in HIV-infected patients with serum sampling for 24 hours after dosing, the observed mean elimination half-life (t ½) ranged from 5 to 7 hours. Oral clearance was 0.37 ± 0.05 L/hr*kg (mean ± SD). Oral clearance and elimination half-life were independent of dose and body weight over an oral dosing range from 0.25 to 10 mg/kg.

Zidovudine

Zidovudine is well absorbed from the gut and, at all dose levels studied, the bioavailability was 60-70%.

Renal clearance of zidovudine greatly exceeds creatinine clearance, indicating that significant tubular secretion takes place.

Zidovudine is primarily eliminated by hepatic conjugation to an inactive glucoronidated metabolite. The 5'-glucuronide of zidovudine is the major metabolite in both plasma and urine, accounting for approximately 50-80% of the administered dose eliminated by renal excretion. 3'-amino-3'-deoxythymidine (AMT) has been identified as a metabolite of zidovudine following intravenous dosing.

Nevirapine

Absorption and Bioavailability

Nevirapine is readily absorbed (>90%) after oral administration in healthy volunteers and in adults with HIV-1 infection. Absolute bioavailability in 12 healthy adults following single-dose administration was 93 ± 9% (mean ± SD) for a 50 mg tablet and 91 ± 8% for an oral solution. Peak plasma nevirapine concentrations of 2 ± 0.4 µg/mL (7.5 µM) were attained by 4 hours following a single 200 mg dose. Following multiple doses, nevirapine peak concentrations appear to increase linearly in the dose range of 200 to 400 mg/day. Steady state trough nevirapine concentrations of 4.5 ± 1.9 µg/mL (17 ± 7 µM), (n = 242) were attained at 400 mg/day. When nevirapine (200 mg) was administered to 24 healthy adults (12 female, 12 male), with either a high fat breakfast (857 kcal, 50 g fat, 53% of calories from fat) or antacid (Maalox® 30 mL), the extent of nevirapine absorption (AUC) was comparable to that observed under fasting conditions.

In a separate study in HIV-1-infected patients (n=6), nevirapine steady-state systemic exposure (AUC[tgr]) was not significantly altered by ddI, which is formulated with an alkaline buffering agent. Nevirapine may be administered with or without food, antacid or ddI.

Distribution

Nevirapine is highly lipophilic and is essentially non-ionized at physiologic pH. Following intravenous administration to healthy adults, the apparent volume of distribution (Vdss) of nevirapine was 1.21 ± 0.09 L/kg, suggesting that nevirapine is widely distributed in humans. Nevirapine readily crosses the placenta and is found in breast milk. Nevirapine is about 60% bound to plasma proteins in the plasma concentration range of 1-10 μg/mL. Nevirapine concentrations in human cerebrospinal fluid (n=6) were 45% (± 5%) of the concentrations in plasma; this ratio is approximately equal to the fraction not bound to plasma protein.

Metabolism/Elimination

In vivo studies in humans and in vitro studies with human liver microsomes have shown that nevirapine is extensively biotransformed via cytochrome P450 (oxidative) metabolism to several hydroxylated metabolites. In vitro studies with human liver microsomes suggest that oxidative metabolism of nevirapine is mediated primarily by cytochrome P450 isozymes from the CYP3A family, although other isozymes may have a secondary role. In a mass balance/excretion study in eight healthy male volunteers dosed to steady state with nevirapine 200 mg given twice daily followed by a single 50 mg dose of 14 C-Nevirapine, approximately 91.4 ± 10.5% of the radiolabeled dose was recovered, with urine (81.3 ± 11.1%) representing the primary route of excretion compared to feces (10.1 ± 1.5%). Greater than 80% of the radioactivity in urine was made up of glucuronide conjugates of hydroxylated metabolites. Thus cytochrome P450 metabolism, glucuronide conjugation, and urinary excretion of glucuronidated metabolites represent the primary route of nevirapine biotransformation and elimination in humans. Only a small fraction (<5%) of the radioactivity in urine (representing <3% of the total dose) was made up of parent compound; therefore, renal excretion plays a minor role in elimination of the parent compound.

Nevirapine has been shown to be an inducer of hepatic cytochrome P450 metabolic enzymes. The pharmacokinetics of autoinduction are characterized by an approximately 1.5 to 2 fold increase in the apparent oral clearance of nevirapine as treatment continues from a single dose to two-to-four weeks of dosing with 200 - 400 mg/day. Autoinduction also results in a corresponding decrease in the terminal phase half-life of nevirapine in plasma from approximately 45 hours (single dose) to approximately 25 - 30 hours following multiple dosing with 200 - 400 mg/day.

Indication

Lamivudine, Nevirapine, and Zidovudine Tablets are indicated alone or in combination with other antiretroviral for the treatment of HIV-1 infection.

Recommended Dose

Therapy should be prescribed by a physician experienced in the management of HIV infection.

The recommended dose of lamivudine, zidovudine and nevirapine tablets in adults is one tablet twice daily.

Lamivudine, zidovudine and nevirapine tablets can be taken with or without food.

Lamivudine, zidovudine and nevirapine tablets should not be administered to patients who have just initiated therapy with nevirapine. This is because an initial lead-in dosing of 200 mg nevirapine once daily for 2 weeks is recommended. Following this lead-in dose, a dose escalation (maintenance dose) to 200mg nevirapine twice-a-day may be carried out in the absence of severe gastrointestinal intolerance or any hypersensitivity reactions (e.g. rash, liver function test abnormalities).

Monitoring of patients

Clinical chemistry tests, which include liver function tests, should be performed prior to initiating lead-in nevirapine therapy and at appropriate intervals during therapy.

Dosage adjustments in patients with haematological adverse reactions:

Dosage reduction or interruption of zidovudine therapy may be necessary in patients whose haemoglobin level falls to between 7.5 g/dl (4.65 mmol/l) and 9 g/dl (5.59 mmol/l) or whose neutrophil count falls to between 0.75 x 10^9/l and 1.0 x 10^9/l. Since lamivudine, zidovudine and nevirapine tablets are a fixed-dose combination, they should not be prescribed for these patient populations.

Renal dysfunction

The dose of lamvudine should be reduced for patients with CrCL <50 mL/min. For zidovudine and nevirapine, dose reduction is required only for severe renal impairment. Nevirapine is dialysed but additional dose is not required if the drug is taken after the dialysis. Since lamivudine, zidovudine and nevirapine tablets are a fixed-dose combination, they should not be prescribed for these patient populations.

Hepatic dysfunction

No dose adjustment for lamivudine is required for patients with impaired hepatic function. However, safety and efficacy of lamivudine has not been established in the presence of decompensated liver disease.

Data in patients with cirrhosis suggest that accumulation of zidovudine may occur in patients with hepatic impairment because of decreased glucuronidation. Dosage reductions may be necessary but, as there is only limited data available, precise recommendations cannot be made.

Increased nevirapine levels and nevirapine accumulation may be observed in patients with serious liver disease. Nevirapine should not be administered to patients with severe hepatic impairment.

Since lamivudine, zidovudine and nevirapine tablets are a fixed-dose combination, they should not be administered in patients with severe hepatic impairment. Caution should be exercised in patients with mild to moderate hepatic impairment.

Dosage in the elderly:

Pharmacokinetics of lamivduine, zidovudine and nevirapine tablets have not been studied in patients over 65 years of age and no specific data are available. However, since special care is advised in this age group due to age-associated changes such as the decrease in renal function and alterations in haematological parameters, appropriate monitoring of patients before and during use of lamivduine, zidovudine and nevirapine tablets is advised.

Lamivudine, zidovudine and nevirapine tablets should be discontinued if patients experience severe rash or a rash accompanied by constitutional findings. Patients experiencing mild to moderate rash during the 14-day lead-in-period of 200 mg/day should not have their nevirapine dose increased or start therapy with lamivudine, zidovudine and nevirapine tablets until the rash has resolved.

Lamivudine, zidovudine and nevirapine tablets administration should be interrupted in patients experiencing severe liver function tests abnormalities (excluding GGT), until the liver function test elevations have returned to baseline. Nevirapine tablets may be restarted at 200 mg per day,

together with lamivudine and zidovudine. Increasing the daily dose to 200 mg twice daily (using lamivudine, zidovudine and nevirapine tablets) should be done with caution, after extended observation. Nevirapine should be permanently discontinued if severe liver function test abnormalities recur.
Patients who interrupt nevirapine dosing for more than 7 days should restart the recommended dosing, using one 200 mg nevirapine tablet daily for the first 14 days (lead-in) in combination with the other antiretrovirals, followed by 200 mg twice daily using lamivudine, zidovudine and nevirapine tablets in the absence of any signs of hypersensitivity.

Route of administration
Oral

Contraindications

Lamivudine, zidovudine and nevirapine tablets are contraindicated in patients with clinically significant hypersensitivity to any of the components contained in the formulation.
Lamivudine, zidovudine and nevirapine tablets are also contraindicated for patients who are just initiating therapy with nevirapine. These patients require a lead-in dose of nevirapine 200 mg once a day, whereas this formulation contains the maintenance dose of nevirapine 200 mg twice-a-day.
Lamivudine, zidovudine and nevirapine tablets should not be given to patients with abnormally low neutrophil counts (less than 0.75×10^9/litre) or abnormally low haemoglobin levels (less than 7.5 g/decilitre or 4.65 mmol/litre).

Warnings and Precautions
The special warnings and precautions relevant to lamivudine, zidovudine and nevirapine are included in this section. There are no additional precautions or warnings relevant to the combination lamivudine, zidovudine and nevirapine tablets.
Skin Reactions
Nevirapine therapy must be initiated with a 14-day lead-in period of 200 mg/day, which has been shown to reduce the frequency of rash. If rash is observed during this lead-in period, dose escalation and administration of lamivudine, zidovudine and nevirapine tablets should not be done until the rash has resolved. Patients should be monitored closely if isolated rash of any severity occurs.
Severe, life-threatening skin reactions, including fatal cases, have occurred in patients treated with nevirapine, occurring most frequently during the first 6 weeks of therapy. These have included cases of Stevens-Johnson syndrome, toxic epidermal necrolysis and hypersensitivity reactions characterized by rash, constitutional findings, and organ dysfunction.
<u>Management</u>
Patients developing signs or symptoms of severe skin reactions or hypersensitivity reactions (including, but not limited to severe rash or rash accompanied by fever, blisters, oral lesions, conjunctivitis, facial edema and or hepatitis, eosinophilia, granulocytopenia, lymphadenopathy and renal dysfunction) must permanently discontinue nevirapine and seek medical evaluation immediately. Nevirapine should not be restarted following severe skin rash or hypersensitivity reaction. Some of the risk factors for developing serious cutaneous reactions include failure to

follow the initial dosing of 200 mg daily during the 14-day lead-in period and delay in stopping nevirapine therapy after the onset of the initial symptoms. If patients present with a suspected nevirapine-associated rash, liver function tests should be performed. Patients with rash-associated SGOT or SGPT elevations should be permanently discontinued for nevirapine.

Women, particularly with higher CD4 cell counts appear to be at higher risk of developing a rash with nevirapine than men. The use of prednisolone to prevent nevirapine-associated rash is not recommended.

Hepatic Events

Nevirapine

Severe, life-threatening and in some cases fatal hepatotoxicity, including fulminant and cholestatic hepatitis, hepatic necrosis and hepatic failure, have been reported in patients treated with nevirapine. The risk is greatest during the first 6 weeks of therapy, and despite significant reduction continues through 18 weeks of treatment. However, hepatic events may occur at any time during treatment. In some cases, patients presented with non-specific prodromal signs or symptoms of fatigue, malaise, anorexia, nausea, jaundice, liver tenderness or hepatomegaly, with or without initially abnormal serum transaminase levels. Some of these events have progressed to hepatic failure with transaminase elevation, with or without hyperbilirubinaemia, prolonged partial thromboplastin time, or eosinophilia. Rash and fever accompanied some of these hepatic events. Patients with signs or symptoms of hepatitis must be advised to discontinue lamivudine, zidovudine and nevirapine tablets and immediately seek medical evaluation, which should include liver function tests.

Women with CD4 counts higher than 250 cells/mm^3 and men with CD4 higher than 400/mm^3 are at greatest risk of hepatic events than more immuno-supressed patients. Patients with significant altered liver enzymes (ALT or AST) and/or clinical hepatitis B or C at the start of therapy can have a greater risk of hepatic adverse events and should be carefully monitored.

Intensive clinical and laboratory monitoring, including liver function tests is essential at baseline and during the first 6 weeks of treatment. Monitoring should continue at frequent intervals thereafter. Liver function tests should be performed immediately if a patient experiences signs or symptoms suggestive of hepatitis and/or hypersensitivity reaction. Liver function tests should also be obtained for all patients who develop a rash in the first 6 weeks of treatment. Physicians and patients should be vigilant for the appearance of signs or symptoms of hepatitis, such as fatigue, malaise, anorexia, nausea, jaundice, bilirubinuria, acholic stools, liver tenderness or hepatomegaly. The diagnosis of hepatotoxicity should be considered in this setting, even if liver function tests are initially normal or alternative diagnoses are possible.

Management

If clinical hepatitis occurs, lamivudine, zidovudine and nevirapine tablets should be permanently discontinued and not restarted after recovery. In some cases, hepatic injury, particularly fibrosis, can progresses despite discontinuation of treatment.

Lamivudine and zidovudine

Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of nucleoside analogues alone or in combination, including lamivudine and zidovudine. Female gender, obesity and prolonged nucleoside exposure may be the major risk factors.

Particular caution should be exercised when administering lamivudine and zidovudine to any patient with known risk factors for liver disease; however, cases have also been reported in patients with no known risk factors. Generalized fatigue, digestive symptoms (nausea, vomiting, abdominal pain and sudden unexplained weight loss); respiratory symptoms (tachypnoea and dyspnoea); or neurologic symptoms (including motor weakness) might be indicative of lactic acidosis development.

Treatment with lamivudine and zidovudine should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis or pronounced hepatoxicity (which may include hepatomegaly and steatosis even in the absence of marked transaminase elevations).

Patients With HIV And Hepatitis B Virus Co-infection

Safety and efficacy of lamivudine has not been established for treatment of chronic hepatitis B in patients dually infected with HIV and HBV. In non-HIV-infected patients treated with lamivudine for chronic hepatitis B, emergence of lamivudine-resistant HBV has been detected and has been associated with diminished treatment response. Emergence of hepatitis B virus variants associated with resistance to lamivudine has also been reported in HIV-infected patients who have received lamivudine-containing antiretroviral regimens in the presence of concurrent infections with hepatitis B virus. Post treatment exacerbations of hepatitis have also been reported.

Post-Treatment Exacerbations Of Hepatitis

In clinical trials in non-HIV-infected patients treated with lamivudine for chronic hepatitis B, clinical and laboratory evidence of exacerbations of hepatitis have occurred after discontinuation of lamivudine. These exacerbations have been detected primarily by serum ALT (STGP) elevations in addition to re-emergence of HBV and DNA. Although most events appear to have been self limited, fatalities have been reported in some cases. Similar events have been reported from post-marketing experience after changes from lamivudine-containing HIV treatment regimens to non-lamivudine containing regimens in patients infected with both HIV and HBV. The causal relationship to discontinuation of lamivudine treatment is unknown. Patients should be closely monitored with both clinical and laboratory follow up for at least several months after stopping treatment. There is insufficient evidence to determine whether re-initiation of lamivudine alters the course of post treatment exacerbations of hepatitis.

Fat Redistribution (Lipodystrophy Syndrome)

Redistribution/accumulation of body fat including central obesity, dorsocervical fat enlargement (buffalo hump), peripheral wasting, facial wasting, breast enlargement and “cushingoid appearance” have been observed in patients receiving antiretroviral therapy. The mechanism and long-term consequences of these events are currently unknown. A causal relationship has not been established.

Haematological Adverse Reactions

Anaemia (usually not observed before six weeks of zidovudine therapy but occasionally occurring earlier), neutropenia (usually not observed before four weeks' therapy but sometimes occurring earlier) and leucopenia (usually secondary to neutropenia) can be expected to occur in patients receiving zidovudine. These occurred more frequently at higher dosages (1200-1500mg/day) and in patients with poor bone marrow reserve prior to treatment, particularly with advanced HIV disease.

Haematological parameters should be carefully monitored. For patients with advanced symptomatic HIV disease it is generally recommended that blood tests are performed at least every two weeks for the first three months of therapy and at least monthly thereafter. In patients with early HIV disease (where bone marrow reserve is generally good), haematological adverse reactions are infrequent. Depending on the overall condition of the patient, blood tests may be performed less often, for example every 1 to 3 months.

If the haemoglobin level falls to between 7.5 g/dl (4.65 mmol/l) and 9 g/dl (5.59 mmol/l) or the neutrophil count falls to between 0.75 x 10^9/l and 1.0 x 10^9/l, the daily dosage may be reduced until there is evidence of marrow recovery; alternatively, recovery may be enhanced by brief (2-4 weeks) interruption of zidovudine therapy. Marrow recovery is usually observed within 2 weeks after which time zidovudine therapy at a reduced dosage may be reinstituted. In patients with significant anaemia, dosage adjustments do not necessarily eliminate the need for transfusions. Since lamivudine, zidovudine and nevirapine tablets are a fixed dose combination product, it will not be possible to reduce the dose and hence in such conditions the patient must be switched over to single ingredient formulations of lamivudine, zidovudine and nevirapine.

Others

The duration of clinical benefit from antiretroviral therapy may be limited. Patients receiving nevirapine or any other antiretroviral therapy may continue to develop opportunistic infection and other complications of HIV infection, and therefore should remain under close clinical observation by physicians experienced in the treatment of patients with associated HIV diseases.

When administering nevirapine as part of an antiretroviral regimen, the complete product information for each therapeutic component should be consulted before initiation of treatment.

Interactions with Other Medicaments

As Lamivudine, zidovudine and nevirapine tablets contains lamivudine, zidovudine and nevirapine, any interactions that have been identified with these agents individually may occur with Lamivudine, zidovudine and nevirapine tablets. The interactions listed below should not be considered exhaustive but are representative of the classes of medicinal products where caution should be exercised.

Interactions relevant to lamivudine

The likelihood of metabolic interactions with lamivudine is low due to limited metabolism and plasma protein binding, and almost complete renal elimination of unchanged lamivudine.

Lamivudine is predominantly eliminated by active organic cationic secretion. The possibility of interactions with other medicinal products administered concurrently should be considered, particularly when their main route of elimination is active renal secretion via the organic cationic transport system e.g.trimethoprim. Other active substances (e.g. ranitidine, cimetidine) are eliminated only in part by this mechanism and were shown not to interact with lamivudine.

Active substances shown to be predominantly excreted either via the active organic anionic pathway, or by glomerular filtration are unlikely to yield clinically significant interactions with lamivudine.

Trimethoprim: Administration of trimethoprim/sulphamethoxazole 160 mg/800 mg (co-trimoxazole) causes a 40% increase in lamivudine exposure because of the trimethoprim component. However, unless the patient has renal impairment, no dosage adjustment of lamivudine is necessary. Lamivudine has no effect on the pharmacokinetics of trimethoprim or sulphamethoxazole. The effect of co-administration of lamivudine with higher doses of co-trimoxazole used for the treatment of *Pneumocystis jiroveci* (*P.carinii*) pneumonia and toxoplasmosis has not been studied.

Zalcitabine: Lamivudine may inhibit the intracellular phosphorylation of Zalcitabine when the two medicinal products are used concurrently. Lamivudine, zidovudine and nevirapine tablets is therefore not recommended to be used in combination with zalcitabine.

Interactions relevant to zidovudine

Zidovudine is primarily eliminated by hepatic conjugation to an inactive glucuronidated metabolite. Active substances which are primarily eliminated by hepatic metabolism especially via glucuronidation may have the potential to inhibit metabolism of zidovudine.

Atovaquone: Zidovudine does not appear to affect the pharmacokinetics of atovaquone. However, pharmacokinetic data shown that atovaquone appears to decrease the rate of metabolism of zidovudine to its glucuronide metabolite (steady state AUC of zidovudine was increased by 33% and peak plasma concentration of the glucuronide was decreased by 19%). At zidovudine dosages of 500 or 600 mg/day it would seem unlikely that a three week, concomitant course of atovaquone for the treatment of acute PCP would result in an increased incidence of adverse reactions attributable to higher plasma concentrations of zidovudine. Extra care should be taken in monitoring patients receiving prolonged atovaquone therapy.

Clarithromycin: Clarithromycin tablets reduce the absorption of zidovudine. This can be avoided by separating the administration of zidovudine and clarithromycin by at least two hours.

Lamivudine: Co-administration of zidovudine with lamivudine results in a 13% increase in zidovudine exposure and a 28% increase in peak plasma levels. However overall exposure (AUC) is not significantly altered. Zidovudine has no effect on the pharmacokinetics of lamivudine.

Phenytoin: Phenytoin blood levels have been reported to be low in some patients receiving zidovudine, while in one patient a high level was noted. These observations suggest that phenytoin concentrations should be carefully monitored in patients receiving Lamivudine, zidovudine and nevirapine tablets and phenytoin.

Probenecid: Limited data suggest that probenecid increases the mean half-life and AUC of zidovudine by decreasing glucuronidation. Renal excretion of the glucuronide (and possibly zidovudine itself) is reduced in the presence of probenecid.

Rifampicin: Limited data suggests that co-administration of zidovudine and rifampicin decreases AUC of zidovudine by 48% ± 34% however the clinical significance of this is unknown.

Stavudine: Zidovudine may inhibit the intracellular phosphorylation of stavudine when the two medicinal products are used concurrently. Stavudine is therefore not recommended to be used in combination with Lamivudine, zidovudine and nevirapine tablets.

Miscellaneous: Other medicinal products, including but not limited to, aspirin, codeine, morphine, methadone, indomethacin, ketoprofen, naproxen, oxazepam, lorazepam, cimetidine, clofibrate, dapsone and Isoprinosine, may alter the metabolism of zidovudine by competitively inhibiting glucuronidation or directly inhibiting hepatic microsomal metabolism. Careful thought should be given to the possibilities of interactions before using such medicinal products particularly for chronic therapy, in combination with Lamivudine, zidovudine and nevirapine tablets.

Concomitant treatment, especially acute therapy, with potentially nephrotoxic or myelosuppressive medicinal products (for example systemic pentamidine, dapsone, pyrimethamine, co-trimoxazole, amphotericin, flucytosine, ganciclovir, interferon, vincristine, vinblastine and doxorubicin) may also increase the risk of adverse reactions to zidovudine. If concomitant therapy with Lamivudine, zidovudine and nevirapine tablets and any of these medicinal products is necessary then extra care should be taken in monitoring renal function and haematological parameters and, if required, the dosage of one or more agents should be reduced.

Since some patients receiving Lamivudine, zidovudine and nevirapine tablets may continue to experience opportunistic infections, concomitant use of prophylactic antimicrobial therapy may have to be considered. Such prophylaxis has included co-trimoxazole, aerosolized pentamidine, pyrimethamine and acyclovir.

Interactions relevant to nevirapine

Other medicinal products metabolized by CYP3A: The induction of CYP3A by nevirapine may result in lower plasma concentrations of other concomitantly administered drugs that are extensively metabolized by CYP3A. Thus, if patients has been stabilized on a dosage regimen for a drug metabolized, by CYP3A, and begins treatment with nevirapine, dose adjustments may be necessary.

Ketoconazole: Nevirapine and ketoconazole should not be administered concomitantly. Coadministration of nevirapine and ketoconazole results in a significant reduction in ketoconazole plasma concentrations.

Fluconazole: Co-administration of fluconazole and nevirapine resulted in approximately 100% increase in nevirapine exposure compared with historical data where nevirapine was administered alone. Because of the risk of increased exposure to nevirapine, caution should be exercised if the medicinal products are given concomitantly and patients should be monitored closely. There was no clinically relevant effect of nevirapine on fluconazole.

Warfarin: The interaction between nevirapine and the antithrombotic agent warfarin is complex, with the potential for both increases and decreases in coagulation time when used concomitantly.

The net effect of the interaction may change during the first weeks of co-administration or upon discontinuation of nevirapine, and close monitoring of anticoagulation levels is therefore warranted.

Oral contraceptives: There are no clinical data on the effects of nevirapine on the pharmacokinetics of oral concentraceptives. Nevirapine may decrease plasma concentrations of oral contraceptives (also other hormonal contraceptives); therefore, these drugs should not be administered concomitantly with nevirapine.

Methadone: based on the known metabolism of methadone, nevirapine may decrease plasma concentrations of methadone by increasing its hepatic metabolism. Narcotic withdrawal syndrome has been reported in patients treated with nevirapine and methadone concomitantly. Methadone-maintained patients beginning nevirapine therapy should be monitored for evidence of withdrawal and methadone dose should be adjusted accordingly.

Side effects

Like all medicines, lamivudine, zidovudine and nevirapine tablets can have side effects. When treating HIV infection, it is not always possible to tell whether some of the side effects that occur are caused by lamivudine, zidovudine and nevirapine tablets, by other medications you are taking at the same time or by the HIV disease. For this reason it is very important that you inform your doctor about any changes in your health. Do not be alarmed by this list of possible side effects of lamivudine, zidovudine and nevirapine tablets, you may not experience them.

Serious side effects that may occur with the use of lamivudine, zidovudine and nevirapine tablets are:

- Lactic acidosis (severe increase of lactic acid in the blood), severe liver enlargement, including inflammation (pain and swelling) of liver and liver failure, which can cause death.
- Symptoms of lactic acidosis may include:
  - Nausea, vomiting, or unusual or unexpected stomach discomfort
  - Feeling very weak and tired
  - Shortness of breath
  - Weakness in arms and legs
- Reduction in production of red blood cells, resulting in anaemia. If this happens, the symptoms are tiredness and shortness of breath.
- Headache, dizziness, pins and needles, inability to concentrate, seizures, depression, feeling anxious and not being able to sleep.
- Breathlessness and cough.
- Nail and skin colour changes, rash (red, raised or itchy) and sweating.
- Liver disorders e.g. enlarged liver, fatty liver and jaundice.
- Inflammation of the pancreas which makes digestive juices and insulin.
- Muscular aches may develop.
- “Flu-like” feeling, fever, tiredness, chills, chest pain, enlarged breasts in male patients, taste changes, general aches and pains, and passing urine more frequently.
- Cardiomyopathy (disease of the heart muscle). Symptoms of this may include shortness of breath, swelling of your ankles or fluid in your lungs.

- Pancreatitis is a dangerous inflammation of the pancreas. It may cause death. Tell your doctor right away if you develop stomach pain, nausea or vomiting. These can be signs of pancreatitis.
- Loss of fat from the legs, arms and face.
- Severe, life-threatening, and in some cases fatal liver disease (hepatitis, liver failure) and skin reactions have been reported. In a small number of patients, rash has been severe and has resulted in death.

Combination antiretroviral therapy may also cause raised sugar in the blood, hyperlipaemia (increased fats in the blood) and resistance to insulin.

Pregnancy and Lactation

Pregnancy

There are no adequate and well-controlled studies in pregnant women for recommending the use of lamivudine, zidovudine and nevirapine tablets. Hence lamivudine, zidovudine and nevirapine tablets should be used during pregnancy only if the potential benefits outweigh the potential risk.

Lactation

It is recommended that HIV-infected mothers do not breast-feed their infants to avoid risking postnatal transmission of HIV infection. It is not known whether lamivudine is excreted in human milk. Zidovudine and nevirapine are present in breast milk.

Symptoms and Treatment of Overdose

There is no experience of overdose with lamivudine, zidovudine and nevirapine tablets. Data reported on the individual components is listed below:

Lamivudine

There is no known antidote for lamivudine. It is not known whether lamivudine can be removed by peritoneal dialysis or hemodialysis.

Zidovudine

Haemodialysis and peritoneal dialysis appear to have a limited effect on elimination of zidovudine but enhance the elimination of the glucuronide metabolite.

Nevirapine

There is no known antidote for nevirapine over -dosage.

Storage Condition

Do not store above 30°C. Keep the container tightly closed. Store in the original package.

Shelf Life

24 months

Presentation

60’s tablets will be packed in HDPE 140ml container along with package insert.

Manufactured by:
Aurobindo Pharma Limited
Unit III, Survey No. 313 & 314,
Bachupally, Bachupally Mandal, Medchal-
Malkajgiri District, Telangana State,
INDIA.

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